

Smart Helmet-Based Bike Safety System

The *Smart Helmet-Based Bike Safety System* is an embedded safety solution designed to enhance the protection of two-wheeler riders by ensuring proper helmet usage and providing real-time alerts during unsafe conditions.

This project focuses on preventing accidents caused due to negligence in wearing helmets. The system is designed such that the bike ignition will only start when the rider is wearing the helmet. A sensor installed inside the helmet detects whether it is properly worn, and this information is wirelessly transmitted to the bike's control unit.

Unlike traditional smart helmet systems, this project removes alcohol detection to simplify the design and reduce cost while focusing on core safety features.

An additional improvement in this system is the integration of **dual alert mechanisms (buzzer + LED)**. If the rider removes the helmet while the bike is running, the system immediately triggers both a buzzer sound and LED indication to alert the rider, thereby preventing unsafe riding behavior.

The system consists of two main modules:

- **Helmet Unit (Transmitter):** Detects helmet usage and sends signals.
- **Bike Unit (Receiver):** Controls ignition and activates alerts based on received signals.

Overall, the project provides a **low-cost, reliable, and practical safety solution** that encourages helmet usage and reduces the risk of severe injuries in road accidents.

Blueprint / Pipeline

Project Flow (Working Pipeline)

Step 1: Helmet Detection

- A **switch / pressure sensor** inside the helmet detects whether it is worn.
- If helmet = NOT worn → signal = 0 → bike ignition OFF.

Step 2: Wireless Communication

- Helmet unit sends signal via **RF Transmitter (433 MHz)**.

- Bike unit receives via **RF Receiver**.

Step 3: Ignition Control

- Microcontroller checks:

```
IF (helmet_worn == TRUE)
  ignition = ON
ELSE
  ignition = OFF
```

Step 4: Real-Time Monitoring (NEW FEATURE)

- If rider removes helmet while riding:
 - **Buzzer ON**
 - **LED ON (visual alert)**

Step 5: Continuous Loop

- System keeps checking helmet status in real time.

System Architecture

Two Modules

♦ Helmet Unit (Transmitter)

- Helmet switch (detect wearing)
- RF transmitter
- Battery

♦ Bike Unit (Receiver)

- RF receiver
- Microcontroller (Arduino / ATmega328)
- Relay (controls ignition)
- Buzzer + LED

Software Requirements

Programming

- Language: **Embedded C / Arduino IDE (C++)**
- Platform: Arduino IDE

Logic Used

- Digital input reading
- Conditional statements (if-else)
- Output control (relay, buzzer, LED)

Hardware Requirements + Budget + Design

Components List

Component	Quantity	Approx Cost (₹)
Arduino Uno / ATmega328	1	250 – 400
RF Transmitter (433 MHz)	1	80 – 120
RF Receiver (433 MHz)	1	80 – 120
Helmet Switch / Push Button	1	20
Relay Module	1	80 – 120
Buzzer	1	20

LED + Resistor	2-3	20
Battery (9V / Li-ion)	1	100 – 200
Connecting Wires	-	50
Breadboard / PCB	1	100

 **Total Estimated Cost: ₹800 – ₹1200**

Design Overview

Helmet Side

- Switch placed inside helmet padding
- RF transmitter attached externally
- Powered by small battery

Bike Side

- Arduino + RF receiver mounted near ignition
- Relay connected to ignition system
- Buzzer + LED mounted on dashboard

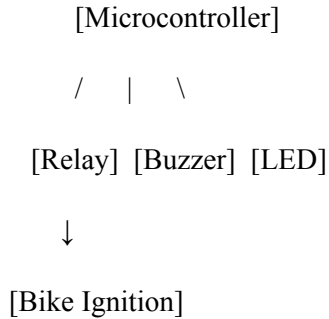
Simple Block Diagram

[Helmet Switch]



[RF Transmitter] ~~~wireless~~~> [RF Receiver]





Baseline Work / References

Inspiration Source

- Smart Helmet Project (Circuits Bazaar)
- Features adapted:
 - Helmet detection
 - RF communication
 - Ignition control

Modifications

- ✓ Removed alcohol detection
- ✓ Added **buzzer + LED alert system**
- ✓ Simplified design for low cost and reliability

YouTube References

1. <https://youtu.be/c7UXiaDgewM?si=K59fRVUY9u3XVyat>
2. <https://youtu.be/qjKzucU256A?si=XW9PxxgXloQaQflps>
3. <https://youtu.be/sATdI9TgsTo?si=ENjOHBXSbo2kpPMJ>

HELMET UNIT CODE (TRANSMITTER)

This code:

- Detects helmet wearing using switch
- Sends signal via RF transmitter

Connections (Helmet Side)

- Switch → Pin 2
- RF Transmitter Data → Pin 12

```
// SMART HELMET - TRANSMITTER CODE

int helmetSwitch = 2; // Helmet detection switch
int rfTx = 12;        // RF Transmitter pin

void setup() {
  pinMode(helmetSwitch, INPUT);
  pinMode(rfTx, OUTPUT);
}

void loop() {

  int helmetState = digitalRead(helmetSwitch);

  if (helmetState == LOW) {
    // Helmet WORN
    digitalWrite(rfTx, HIGH);
  } else {
    // Helmet NOT WORN
    digitalWrite(rfTx, LOW);
  }

  delay(100);
}
```

BIKE UNIT CODE (RECEIVER + CONTROL)

👉 This is the main brain:

- Receives signal
- Controls ignition (relay)
- Activates **buzzer + LED** if helmet removed while riding

Connections (Bike Side)

Component	Pin
RF Receiver	Pin 7
Relay (Ignition)	Pin 8
Buzzer	Pin 2
LED	Pin 3

```
#include <LiquidCrystal.h>

// LCD PINS (same as your base code)
LiquidCrystal lcd(3, 13, 9, 10, 11, 12);

int rfRx = 7;    // RF Receiver input
int relay = 8;   // Ignition control
int buzzer = 2;
int led = 4;
```

```
int helmetStatus = 0;

int prevHelmetStatus = 0;


void setup() {

    pinMode(rfRx, INPUT);

    pinMode(relay, OUTPUT);

    pinMode(buzzer, OUTPUT);

    pinMode(led, OUTPUT);


    lcd.begin(16, 2);


    lcd.setCursor(0, 0);

    lcd.print("SMART HELMET");

    lcd.setCursor(0, 1);

    lcd.print("SYSTEM READY");


    delay(2000);

    lcd.clear();

}


void loop() {
```



```
helmetStatus = digitalRead(rfRx);

// ♦ CASE 1: HELMET WORN → START BIKE

if (helmetStatus == HIGH) {

    digitalWrite(relay, LOW); // IGNITION ON

    digitalWrite(buzzer, LOW);

    digitalWrite(led, LOW);


    lcd.setCursor(0, 0);

    lcd.print("HELMET WORN ");

    lcd.setCursor(0, 1);

    lcd.print("IGNITION ON ");

}

// ♦ CASE 2: HELMET NOT WORN BEFORE START

if (helmetStatus == LOW && prevHelmetStatus == LOW) {

    digitalWrite(relay, HIGH); // IGNITION OFF


    lcd.setCursor(0, 0);

    lcd.print("NO HELMET ");

    lcd.setCursor(0, 1);

    lcd.print("BIKE OFF ");

}
```

```
// ♦ CASE 3: HELMET REMOVED WHILE RIDING (IMPORTANT FEATURE)
```

```
if (helmetStatus == LOW && prevHelmetStatus == HIGH) {
```

```
    lcd.clear();
```

```
    lcd.setCursor(0, 0);
```

```
    lcd.print("HELMET REMOVED");
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("ALERT !!!  ");
```

```
    // Activate ALERT
```

```
    for (int i = 0; i < 5; i++) {
```

```
        digitalWrite(buzzer, HIGH);
```

```
        digitalWrite(led, HIGH);
```

```
        delay(300);
```

```
        digitalWrite(buzzer, LOW);
```

```
        digitalWrite(led, LOW);
```

```
        delay(300);
```

```
    }
```

```
    digitalWrite(relay, HIGH); // TURN OFF BIKE
```

```
}
```

```
prevHelmetStatus = helmetStatus;
```

```
delay(200);
```

```
}
```

Circuit Diagram

